Unit 3.1

Ex 3.1 p180

3. 0

5. 5.2

7. $6t^2 - 6t - 4$

9. $2x - 6x^2$

11. $-\frac{2}{5x^{7/5}}$

13. $-\frac{15}{r^4}$

15. $18a + 6$

17. $\frac{1}{2}p^{-1/2}$

19. $3e^x - \frac{4}{3}x^{-4/3}$

21. $3Au^2 + 2Bu + C$

23. $\frac{3}{2}\sqrt{x} + 2/\sqrt{x} - \frac{3}{2x\sqrt{x}}$

25. $2.4x^{1.4}$

27. $G(q) = (1-q^{-1})^2 = 1 - 2q^{-1} + q^{-2}$

$G'(q) = 2q^{-2} - 2q^{-3}$

29. $f(v) = \frac{\sqrt[3]{v} - 2ve^v}{v}$

$= v^{-2/3} - 2e^v$

$f'(v) = -\frac{2}{3}v^{-5/3} - 2e^v$

31. $-10Ay^{-11} + Be^y$

33. $y = 2x^3 - x^2 + 2$

$\Rightarrow y' = 6x^2 - 2x$. At $x=1$

$y' = 4$ and equation is

$y - 3 = 4(x-1) \Rightarrow y = 4x - 1$

36. $y = \sqrt[4]{x} - x$ $(1,0)$

$y' = \frac{1}{4}x^{-3/4} - 1$. At $x=1$

$y' = \frac{1}{4} - 1 = -\frac{3}{4}$ so

equation of tangent line

$y - 0 = -\frac{3}{4}(x-1)$

45. $f(x) = 0.001x^5 - 0.02x^3$

$f'(x) = 0.005x^4 - 0.06x^2$

$f''(x) = 0.020x^3 - 0.12x$

49. (a) $s = t^3 - 3t$

$\Rightarrow v(t) = s'(t) = 3t^2 - 3$ and

$a(t) = v'(t) = 6t$

(b) $a(2) = 12$ ms^{-2}

(c) $v(t) = 3t^2 - 3 = 0$

$\Rightarrow t = 1$ $(t \geq 0)$

and $a(1) = 6$ ms^{-2}

55. $y = 2x^3 + 3x^2 - 12x + 1$

Horizontal tangent when

$y' = 6x^2 + 6x - 12 = 0$

$6(x+2)(x-1) = 0$

$\Rightarrow x = -2$ or $x = 1$ Points on the curve are

$(-2, 21)$ and $(1, -6)$

56. $f(x) = e^x - 2x$

$f'(x) = e^x - 2 = 0$ (for horizontal tangents)

$\Rightarrow e^x = 2$

$\Rightarrow x = \ln 2$

57. $y = 2e^x + 3x + 5x^3$

$\Rightarrow y' = 2e^x + 3 + 15x^2$

Assume the slope is 2 then

$2e^x + 3 + 15x^2 = 2$

$\Rightarrow 2e^x + 15x^2 = -1$

This is not possible since $e^x > 0$ and $x^2 \geq 0$ for all x. Tangent line can not have a slope of 2.

58. $y = x^4 + 1$

$\Rightarrow y' = 4x^3$. The line

$32x - y = 15 \Rightarrow y = 32x - 15$

has a slope of 32.

The line parallel will also have a slope of 32.

$\Rightarrow 4x^3 = 32$

$x^3 = 8 \Rightarrow x = 2$

and the y-coordinate is

$y = (2)^4 + 1 = 17$

Equation of tangent line:

Has slope of 32 passing through $(2,17)$

$y - 17 = 32(x-2)$

$y = 32x - 47$

61. $y = \sqrt{x}$

$\Rightarrow y' = \frac{1}{2\sqrt{x}}$.

The slope of $2x + y = 1$ is -2. so the normal line must have a slope of -2, Hence the tangent line must have a slope of $\frac{1}{2}$.

Therefor $\frac{1}{2\sqrt{x}} = \frac{1}{2}$

$\Rightarrow \sqrt{x} = 1 \Rightarrow x = 1$

If $x=1$, then $y = 1$ so the desired equation has slope of -2 passing through $(1,1)$

$y - 1 = -2(x-1)$

$y = -2x + 3$

74. If $x \geq 1$ then

$h(x) = |x-1| + |x+2|$

$= x - 1 + x + 2 = 2x + 1$

If $-2 \leq x \leq 1$

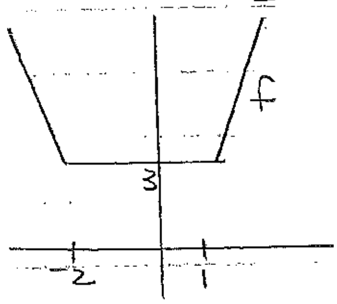
$h(x) = -(x-1) + x + 2 = 3$

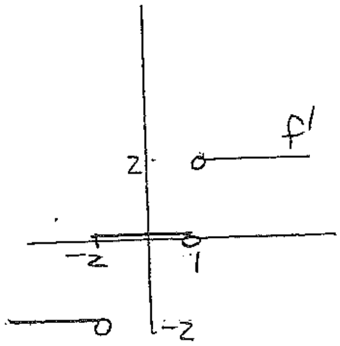
If $x \leq -2$

$h(x) = -(x-1) - (x+2) = -2x - 1$

$$h(x) = \begin{cases} -2x-1 & \text{if } x \leq -2 \\ 3 & \text{if } -2 < x \leq 1 \\ 2x-1 & \text{if } x \geq 1 \end{cases}$$

$$\Rightarrow h'(x) = \begin{cases} -2 & \text{if } x < -2 \\ 0 & \text{if } -2 < x < 1 \\ 2 & \text{if } x > 1 \end{cases}$$





$h'(1)$ and $h'(-2)$ does not exist.

81. $$f(x) = \begin{cases} x^2 & \text{if } x \leq 2 \\ mx + b & \text{if } x > 2 \end{cases}$$

f is clearly differentiable for $x < 2$ and for $x > 2$.

For $x < 2$ $f'(x) = 2x$

$f'_-(2) = 4$

For $x > 2$ $f'(x) = m \Rightarrow m = 4$

$\Rightarrow f(x) = 4x + b$

For continuity at $x = 2$

$f(2) = 4$

$\lim_{x \to 2^-} x^2 = 4$

$\lim_{x \to 2^+} (4x + b) = 8 + b$

So $8 + b = 4$

$\Rightarrow b = -4$

Unit 3.2

Ex 3.2 p 188

31. $y' = \dfrac{x^2+4x+1}{(x^2+x+1)^2}$ At $(1,0)$

$y' = \frac{2}{3}$ $\Rightarrow$ Equation of tangent $y - 0 = \frac{2}{3}(x-1)$

35 (a) $y = \dfrac{1}{1+x^2}$ $\Rightarrow$

$y' = \dfrac{-2x}{(1+x^2)^2}$ At $(-1, \frac{1}{2})$

$y' = \dfrac{2}{4} = \dfrac{1}{2}$ So equation of tangent line

$y - \frac{1}{2} = \frac{1}{2}(x+1)$

$y = \frac{1}{2}x + 1$

41. $f(x) = \dfrac{x^2}{1+x}$

$f'(x) = \dfrac{x^2+2x}{x^2+2x+1}$

$f''(x) = \dfrac{2}{(x+1)^3}$

$\Rightarrow f''(1) = \dfrac{2}{8} = \dfrac{1}{4}$

43. $f(5) = 1$ $f'(5) = 6$

$g(5) = -3$ $g'(5) = 2$

(a) $(fg)'(5) = f'(5)g(5) + f(5)g'(5)$

$= (6)(-3) + (1)(2) = -16$

b) $\left(\frac{f}{g}\right)'(5) = \dfrac{f'(5)g(5) - f(5)g'(5)}{[g(5)]^2}$

$= \dfrac{(6)(-3) - (1)(2)}{(-3)^2}$

$= \dfrac{-18-2}{9} = -\dfrac{20}{9}$

c) $\left(\frac{g}{f}\right)'(5) = 20$

Unit 3.3

21. $y' = \cos x - \sin x$ so

$y'(0) = 1$. Equation is

$y - 1 = 1(x - 0) \Rightarrow y = x + 1$

24. $y' = 1 + \sec^2 x$ so

$y'(\pi) = 1 + (-1)^2 = 2$ $\Rightarrow$

$y - \pi = 2(x - \pi)$

$\Rightarrow y = 2x - \pi$ is eq of tang. line

26 (a) $y' = 3 - 6\sin x$ so

$y'\left(\frac{\pi}{3}\right) = 3 - 6\left(\frac{\sqrt{3}}{2}\right)$ and

equation of tang. line is

$y - (\pi + 3) = (3 - 3\sqrt{3})(x - \frac{\pi}{3})$

$\therefore y = (3 - 3\sqrt{3})x + 3 + \pi\sqrt{3}$

28(a) $f(x) = e^x \cos x$

$f'(x) = e^x \cos x - e^x \sin x$

$= e^x(\cos x - \sin x)$

$f''(x) = e^x(\cos x - \sin x) +$

$e^x(-\sin x - \cos x)$

$= -2e^x \sin x$

29. $H(\theta) = \theta \sin\theta$

$H'(\theta) = \sin\theta + \theta\cos\theta$

$H''(\theta) = \cos\theta + \cos\theta - \theta\sin\theta$

$= 2\cos\theta - \theta\sin\theta$

33. $f(x) = x + 2\sin x$

$\Rightarrow f'(x) = 1 + 2\cos x = 0$

for horizontal tangent

$\Rightarrow \cos x = -\frac{1}{2}$

$\Rightarrow x = \frac{2\pi}{3} + 2\pi n$ or

$x = \frac{4\pi}{3} + 2\pi n$, $n \in \mathbb{Z}$

OR $x = (2n+1)\pi \pm \pi/3$

Unit 3.4 Ex 3.4 p.204

7. $F'(x) = 4(5x^6+2x^3)^3\,[30x^5+6x^2]$

9. $f'(x) = \frac{1}{2}(5x+1)^{-1/2}\,[5]$

11. $f'(\theta) = -\sin(\theta^2)\cdot 2\theta$

13. $y' = 2xe^{-3x} + x^2e^{-3x}\cdot(-3)$

15. $f'(t) = ae^{at}(\sin bt) + \cos bt(b)e^{at}$
$= e^{at}(a\sin bt + b\cos bt)$

17. $f'(x) = (2x-3)^4\cdot 5(x^2+x+1)^4\cdot(2x+1) + (x^2+x+1)^5\cdot 4(2x-3)^3\cdot 2$

19. $h'(x) = \frac{2}{3}(t+1)^{-1/3}(2t^2-2)^2\cdot(20t^2+18t-1)$

21. $y' = \frac{1}{2}\left(\frac{x}{x+1}\right)^{-1/2}\cdot\frac{(x+1)-x}{(x+1)^2}$
$= \dfrac{1}{2\sqrt{x}\,(x+1)^{3/2}}$

52. $y' = \frac{1}{2}(1+x^3)^{-1/2}\cdot 3x^2$ so
$y'\big|_{(2,3)} = \frac{1}{2}(1+8)^{-1/2}\,3(2)^2$
$= \frac{1}{2}\cdot\frac{1}{3}\cdot 3\cdot 4$
$= 2$ So
equation of tangent line is $y-3 = 2(x-2)$
$y = 2x-1$

62. $h(x) = \sqrt{4+3f(x)}$
$h'(x) = \frac{1}{2}(4+3f(x))^{-1/2}\cdot 3f'(x)$
$h'(1) = \frac{1}{2}(4+3f(1))^{-1/2}\cdot 3f'(1)$
$= \frac{1}{2}(4+3\cdot 7)^{-1/2}\cdot 3(4)$
$= \frac{1}{2}\cdot\frac{1}{5}\cdot 12$
$= \frac{6}{5}$

63. (a) $h(x) = f(g(x))$
$\Rightarrow h'(x) = f'(g(x))\cdot g'(x)$
$\Rightarrow h'(1) = f'(g(1))\cdot g'(1)$
$= f'(2)\cdot 6$
$= 5\cdot 6 = 30$

(b) $H(x) = g(f(x))$
$\Rightarrow H'(x) = g'(f(x))\cdot f'(x)$
$\Rightarrow H'(1) = g'(f(1))\cdot f'(1)$
$= g'(3)\cdot 4$
$= 9\cdot 4 = 36$

79. $s(t) = 10 + \frac{1}{4}\sin(10\pi t)$
$\Rightarrow v(t) = \frac{1}{4}\cos(10\pi t)\cdot 10\pi$
$= \frac{5\pi}{2}\cos(10\pi t)$ cm/s

84 (a) $\lim_{t\to\infty} p(t)$
$= \lim_{t\to\infty}\dfrac{1}{1+ae^{-kt}} = \dfrac{1}{1+a\cdot 0}$
$= 1$

(b) $p(t) = (1+ae^{-kt})^{-1}$
$p'(t) = -(1+ae^{-kt})^{-2}\cdot ae^{-kt}\cdot -k$
$= \dfrac{kae^{-kt}}{(1+ae^{-kt})^2}$

96. The rate of change of y^5 with respect to x is 80 times the rate of chang of y with respect to x.
$\frac{d}{dx}y^5 = 80\frac{dy}{dx}$
$5y^4 = 80$ ($\frac{dy}{dx} \neq 0$ — no horizonta tangent)
$\Rightarrow y^4 = 16$
$\Rightarrow y = 2$ $(y>0)$

Unit 3.5

Ex 3.5 p215

5. $\frac{d}{dx}(x^2-4xy+y^2)=\frac{d}{dx}4$

$\Rightarrow 2x-4(x'y+xy')+2yy'=0$

$\Rightarrow 2x-4y-4xy'+2yy'=0$

$\Rightarrow y'[-4x+2y]=-2x+4y$

$y'=\frac{2(2y-x)}{2(y-2x)}=\frac{2y-x}{y-2x}$

7. $4x^3+2xy^2+x^2 2yy'+3y^2y'=0$

$y'[2x^2y+3y^2]=-4x^3-2xy^2$

$y'=-\frac{2x(2x^2+y^2)}{y(2x^2+3y)}$

9. $\frac{2x(x+y)-x^2(1+y')}{(x+y)^2}=2y.y'$

$\Rightarrow 2x^2+2xy-x^2-x^2y'=2yy'(x+y)^2$

$y'[-x^2-2y(x+y)^2]=-x^2-2xy$

$y'=\frac{x(x+2y)}{x^2+2y(x+y)^2}$

11. $2xy^2+x^2 2yy'+\sin y+x\cos y.y'=0$

$\Rightarrow y'[2x^2y+x\cos y]=-2xy^2-\sin y$

$y'=\frac{-2xy^2-\sin y}{2x^2y+x\cos y}$

13. $\frac{1}{2}(x+y)^{-1/2}(1+y')=4x^3+4y^3.y'$

$\frac{1+y'}{2\sqrt{x+y}}=4x^3+4y^3.y'$

$1+y'=8x^3\sqrt{x+y}+8y^3\sqrt{x+y}\,y'$

46. $\sqrt{x}+\sqrt{y}=\sqrt{c}$

$\Rightarrow \frac{1}{2\sqrt{x}}+\frac{y'}{2\sqrt{y}}=0$

$\Rightarrow y'=\frac{\sqrt{y}}{\sqrt{x}}$ and equation of tangent line through a point (x_0,y_0) is.

$y-y_0=-\frac{\sqrt{y_0}}{\sqrt{x_0}}(x-x_0)$

y-intercept where $x=0$

$\Rightarrow y=y_0-\frac{\sqrt{y_0}}{\sqrt{x_0}}(-x_0)$

$=y_0+\sqrt{x_0}\sqrt{y_0}$

$(0,\ y_0+\sqrt{x_0y_0})$ and if $y=0$ then $-y_0=-\frac{\sqrt{y_0}}{\sqrt{x_0}}(x-x_0)$

$\Rightarrow (x-x_0)=-y_0\left(-\frac{\sqrt{x_0}}{\sqrt{y_0}}\right)$

$x=x_0+\sqrt{x_0}\sqrt{y_0}$

$(x_0+\sqrt{x_0y_0},\ 0)$ and the sum of the two intercepts:

$(x_0+\sqrt{x_0y_0})+(y_0+\sqrt{x_0y_0})$

$=x_0+2\sqrt{x_0}\sqrt{y_0}+y_0$

$=(\sqrt{x_0}+\sqrt{y_0})^2=(\sqrt{c})^2=c$

49. $y=\tan^{-1}\sqrt{x}$

$y'=\frac{\frac{1}{2\sqrt{x}}}{1+(\sqrt{x})^2}$

$=\frac{1}{2\sqrt{x}(1+x)}$

50. $y=\sqrt{\tan^{-1}x}=(\tan^{-1}x)^{1/2}$

$y'=\frac{1}{2}(\tan^{-1}x)^{-1/2}\cdot\frac{1}{1+x^2}$

$=\frac{1}{2\sqrt{\tan^{-1}x}\,(1+x^2)}$

54. $y = \tan^{-1}(x - \sqrt{x^2+1})$

$$y' = \frac{1 - \frac{1}{2}(x^2+1)^{-1/2} \cdot 2x}{1 + (x - \sqrt{x^2+1})^2}$$

$$= \frac{1 - \frac{x}{\sqrt{x^2+1}}}{1 + x^2 - 2x\sqrt{x^2+1} + (x^2+1)}$$

$$= \frac{\sqrt{x^2+1} - x}{(1+x^2)(\sqrt{x^2+1} - x)2}$$

$$= \frac{1}{2(1+x^2)}$$

57. $y = x\sin^{-1}x + \sqrt{1-x^2}$

$$y' = \sin^{-1}x + x \cdot \frac{1}{\sqrt{1-x^2}} + \frac{1}{2}(1-x^2)^{-1/2} \cdot (-2x)$$

$$= \sin^{-1}x$$

73. x-intercepts where $y=0$

$\Rightarrow x^2 - x(0) + 0^2 = 3$

$\Rightarrow x = \pm\sqrt{3}$

$x^2 - xy + y^2 = 3$

$\Rightarrow 2x - (y + xy') + 2y \cdot y' = 0$

$\Rightarrow y'(2y - x) = y - 2x$

$y' = \frac{y-2x}{2y-x}$

So y' at $(\sqrt{3}, 0)$ is 2 and y' at $(-\sqrt{3}, 0)$ is 2

$\Rightarrow$ Tangent lines are parallel.

76. $x^2 + 4y^2 = 36$

$\Rightarrow 2x + 8y \cdot y' = 0$

$\Rightarrow y' = -\frac{x}{4y}$

Let (a,b) be a point on $x^2+4y^2=36$ whose tangent line passes through $(12,3)$.

Eq of tangent line is.

$$y - 3 = -\frac{a}{4b}(x - 12)$$

$$\Rightarrow b - 3 = -\frac{a}{4b}(a-12) \quad (1)$$

$\Rightarrow 4b^2 - 12b = -a^2 + 12a$

$\Rightarrow 4b^2 + a^2 = 12(a+b)$

but $4b^2 + a^2 = 36 \Rightarrow$

$36 = 12(a+b)$

$\Rightarrow a + b = 3$

$\Rightarrow b = 3 - a$

Subst this into

$a^2 + 4b^2 = 36$

$\Rightarrow a^2 + 4(3-a)^2 = 36$

$\Rightarrow 5a^2 - 24a = 0$

$\Rightarrow a = 0$ or $a = \frac{24}{5}$

If $a = 0$ then $b = 3$ and $a = \frac{24}{5}$ then $b = -\frac{9}{5}$

So the two points on the ellips are

$(0,3)$ and $(\frac{24}{5}, -\frac{9}{5})$

Substituting into (1) gives the tangent lines:

$y = 3$ and

$y = \frac{2}{3}x - 5.$

Unit 3·6

Ex 3·6 p223

3. $f'(x) = \cos(\ln x) \cdot \frac{1}{x}$

5. $f'(x) = \frac{1}{5}(\ln x)^{-4/5} \cdot \frac{1}{x}$

7. $f'(x) = \frac{3x^2}{(x^3+1)\ln 10}$

9. $f'(x) = \cos x \cdot \ln(5x) + \sin x \cdot \frac{5}{5x}$

11. $f(x) = \ln x + \frac{1}{2}\ln(x^2-1)$

$f'(x) = \frac{1}{x} + \frac{2x}{2(x^2-1)}$

23. $y = \sqrt{x}\,\ln x$

$y' = \frac{1}{2\sqrt{x}}\ln x + \frac{\sqrt{x}}{x}$

$= \frac{1}{2}\frac{\ln x}{\sqrt{x}} + \frac{1}{\sqrt{x}}$

$= \frac{1}{2}\frac{1}{\sqrt{x}}(\ln x + 1)$

$y'' = \frac{1}{2}\left(-\frac{1}{2}x^{-3/2}\right)(\ln x + 1) + \frac{1}{2}x^{-1/2}\left(\frac{1}{x}\right)$

26. $y = \ln(1+\ln x)$

$y' = \frac{\frac{1}{x}}{1+\ln x} = \frac{1}{x(1+\ln x)}$

$y'' = \frac{-[(1+\ln x) + x\cdot\frac{1}{x}]}{x^2(1+\ln x)^2}$

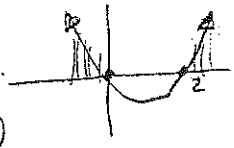
$= \frac{-\ln x}{x^2(1+\ln x)^2}$

29. $f(x) = \ln(x^2-2x)$

$\Rightarrow f'(x) = \frac{2x-2}{x^2-2x}$

f defined where $x^2-2x > 0$

$\Rightarrow x(x-2) > 0$



$D_f = (-\infty, 0) \cup (2, \infty)$

39. $y = (2x+1)^5(x^4-3)^6$

$\Rightarrow \ln y = 5\ln(2x+1) + 6\ln(x^4-3)$

$\Rightarrow \frac{1}{y}\cdot y' = 5\,\frac{2}{2x+1} + 6\,\frac{4x^3}{x^4-3}$

$\Rightarrow y' = y\left[\frac{10}{2x+1} + \frac{24x^3}{x^4-3}\right]$

41. $y = \sqrt{\frac{x-1}{x^4+1}}$

$\Rightarrow \ln y = \frac{1}{2}[\ln(x-1) - \ln(x^4+1)]$

$\Rightarrow \frac{1}{y}y' = \frac{1}{2}\left[\frac{1}{x-1} - \frac{4x^3}{x^4+1}\right]$

$\Rightarrow y' = \frac{1}{2}y\left[\frac{1}{x-1} - \frac{4x^3}{x^4+1}\right]$

45. $y = x^{\sin x}$

$\Rightarrow \ln y = \sin x \ln x$

$\Rightarrow \frac{1}{y}y' = \cos x \ln x + \frac{\sin x}{x}$

$\Rightarrow y' = y\left[\cos x \ln x + \frac{\sin x}{x}\right]$

47. $y = (\cos x)^x$

$\Rightarrow \ln y = x \ln \cos x$

$\Rightarrow \frac{1}{y}y' = \ln\cos x + x\cdot\frac{-\sin x}{\cos x}$

$\Rightarrow y' = y[\ln\cos x - x\tan x]$

51. $y = \ln(x^2+y^2)$

$y' = \frac{2x + 2yy'}{x^2+y^2}$

$\Rightarrow y'(x^2+y^2) - 2yy' = 2x$

$\Rightarrow y'[x^2+y^2-2y] = 2x$

$\Rightarrow y' = \frac{2x}{x^2+y^2-2y}$

Unit 3.7

Ex 3.11 p264

(a) $\sinh 0 = \frac{1}{2}(e^0 - e^0) = 0$

(b) $\cosh 0 = \frac{1}{2}(e^0 + e^0) = 1$

(a) $\tanh 0 = \frac{0}{1} = 0$

(b) $\tanh 1 = \frac{e^1 - e^{-1}}{e^1 + e^{-1}} = \frac{e^2 - 1}{e^2 + 1}$

≈ 0.7616

(a) $\cosh(\ln 5)$

$= \frac{1}{2}(e^{\ln 5} + e^{-\ln 5}) = \frac{1}{2}(5 + \frac{1}{5})$

$= \frac{26}{10} = \frac{13}{5}$

(b) $\cosh 5$

$= \frac{1}{2}(e^5 + e^{-5})$

3. (a) $\lim_{x\to\infty} \tanh x = \lim_{x\to\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}}$

$= \lim_{x\to\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} \cdot \frac{e^{-x}}{e^{-x}}$

$= \lim_{x\to\infty} \frac{1 - e^{-2x}}{1 + e^{-2x}} = 1$

(b) $\lim_{x\to-\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} \cdot \frac{e^x}{e^x}$

$= \lim_{x\to-\infty} \frac{e^{2x} - 1}{e^{2x} + 1} = -1$

(c) $\lim_{x\to\infty} \sinh x = \lim_{x\to\infty} \frac{e^x - e^{-x}}{2}$

$= \infty$

(d) $\lim_{x\to-\infty} \frac{e^x - e^{-x}}{2} = -\infty$

(e) $\lim_{x\to\infty} \operatorname{sech} x = \lim_{x\to\infty} \frac{2}{e^x + e^{-x}}$

$= 0$

(f) $\lim_{x\to\infty} \coth x = \lim_{x\to\infty} \frac{e^x + e^{-x}}{e^x - e^{-x}}$

$= \lim_{x\to\infty} \frac{1 + e^{-2x}}{1 - e^{-2x}} = 1$

(g) $\lim_{x\to 0^+} \coth x = \lim_{x\to 0^+} \frac{\cosh x}{\sinh x} = \infty$

(h) $\lim_{x\to 0^-} \coth x = -\infty$

(i) $\lim_{x\to-\infty} \operatorname{cosech} x = \lim_{x\to-\infty} \frac{2}{e^x - e^{-x}}$

$= 0$

30. $f(x) = e^x \cosh x$

$f'(x) = e^x \cosh x + e^x \sinh x$

$= \frac{1}{2} e^x (e^x + e^{-x} + e^x - e^{-x})$

$= \frac{1}{2} e^x e^{2x} = \frac{1}{2} e^{3x}$

32. $g(x) = \sinh^2 x$

$g'(x) = 2 \sinh x \cdot \cosh x$

$= 2\left(\frac{e^x - e^{-x}}{2}\right)\left(\frac{e^x + e^{-x}}{2}\right)$

$= \frac{1}{2}\left[e^{2x} - e^{-2x}\right]$

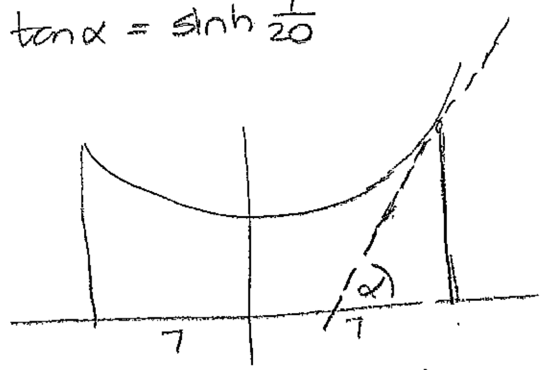
51. $y = 20 \cosh\left(\frac{x}{20}\right) - 15$

(a) $y' = 20 \sinh\left(\frac{x}{20}\right) \cdot \frac{1}{20}$

Right pole is positioned at $x = 7$ therefore:

$y'(7) = \sinh\left(\frac{7}{20}\right) \approx 0.3572$

(b) If α is the angle between the tangent line and the x-axis then $\tan\alpha$ = slope of the line so $\tan\alpha = \sinh\frac{7}{20}$



$\Rightarrow \alpha = \tan^{-1}\left(\sinh\frac{7}{20}\right) \approx 0.343$ rad

$\approx 19.66°$

$\Rightarrow$ Angle between line and pole is $90° - 19.66° = 70.34°$